

Kubernetes Lab: Program 9 - Namespaces and Labels

This document details the configuration and execution steps for managing Kubernetes resources using Namespaces and Labels.

Project Structure

Program-9/

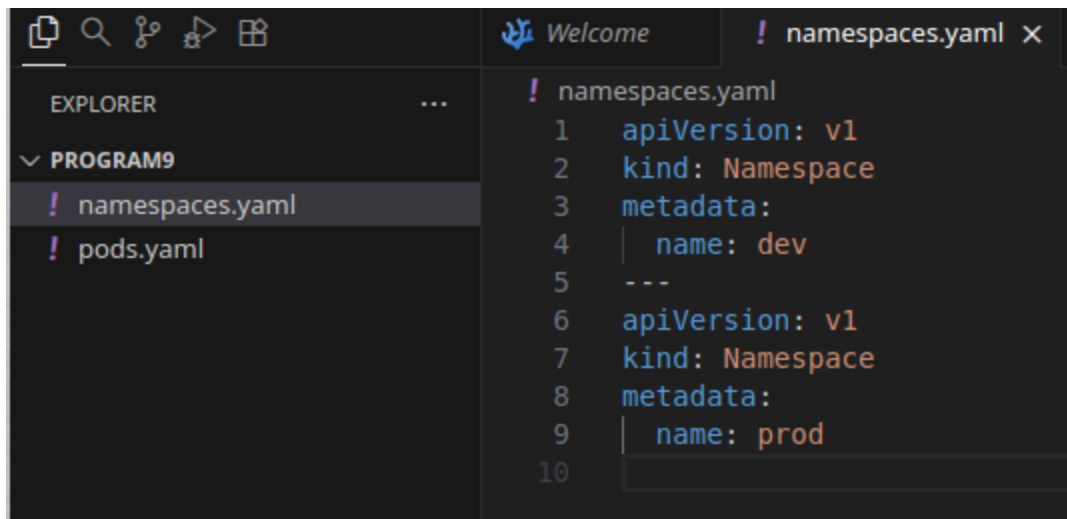
```
|— namespaces.yaml  
|— pods.yaml
```

Configuration Files

1. Namespace Configuration (namespaces.yaml)

This file defines two isolated environments: dev and prod.

```
apiVersion: v1  
kind: Namespace  
metadata:  
  name: dev  
---  
apiVersion: v1  
kind: Namespace  
metadata:  
  name: prod
```



2. Pod Configuration (pods.yaml)

This file defines frontend and backend pods for both namespaces.

Dev Namespace Pods

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

Prod Namespace Pods

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
```

```
namespace: prod
labels:
  app: frontend
  env: prod
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

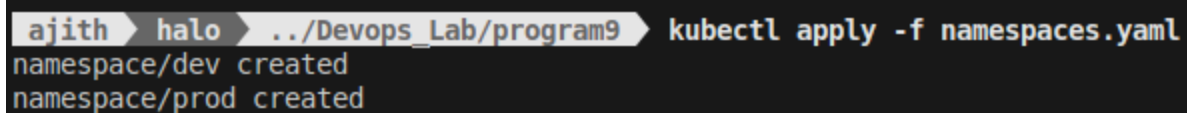
Execution Steps

Step 1: Apply Configuration

Apply the namespaces first, followed by the pods.

```
kubectl apply -f namespaces.yaml
```

```
kubectl apply -f pods.yaml
```

A terminal window with a dark background. The prompt is 'ajith@halo' followed by the directory path ' ../Devops_Lab/program9'. The command 'kubectl apply -f namespaces.yaml' has been executed, resulting in two lines of output: 'namespace/dev created' and 'namespace/prod created'.

```
ajith@halo: ../Devops_Lab/program9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
```

```
ajith@halo:~/Devops_Lab/program9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
```

Step 2: Verify Resources

Check the pods specifically within each namespace.

Check Dev namespace

```
kubectl get pods -n dev
```

Check Prod namespace

```
kubectl get pods -n prod
```

Check all namespaces to see everything

```
kubectl get pods --all-namespaces
```

```
ajith@halo:~/Devops_Lab/program9$ kubectl get pods -n dev
```

NAME	READY	STATUS	RESTARTS	AGE
backend-pod	1/1	Running	0	13s
frontend-pod	1/1	Running	0	13s

```
ajith@halo:~/Devops_Lab/program9$ kubectl get pods -n prod
```

NAME	READY	STATUS	RESTARTS	AGE
backend-pod	1/1	Running	0	25s
frontend-pod	1/1	Running	0	25s

```
ajith@halo:~/Devops_Lab/program9$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
demo-deploy-6876f5fc77-8jdlf	1/1	Running	6 (6d22h ago)	17d
demo-pod1	1/1	Running	5 (6d22h ago)	17d
hello-2s5jb	0/1	Completed	0	17d
hello-775d79c56b-xfkxt	1/1	Running	5 (6d22h ago)	17d
hello2-8797688b4-q662v	1/1	Running	5 (6d22h ago)	17d
nginx-deployment-6f9664446b-kfddw	1/1	Running	1 (6d22h ago)	7d10h
nginx-deployment-6f9664446b-q586r	1/1	Running	1 (6d22h ago)	7d10h
program-8	1/1	Running	1 (2m41s ago)	6d23h
yaml-demo	1/1	Running	4 (6d22h ago)	17d

Step 3: Filter Using Labels

Labels allow you to select specific pods across or within namespaces.

List all frontend pods in the dev namespace

```
kubectl get pods -n dev -l app=frontend
```

List all backend pods in the prod namespace

```
kubectl get pods -n prod -l app=backend
```

```
ajith@halo:~/Devops_Lab/program9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           4m22s
```

```
ajith@halo:~/Devops_Lab/program9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           4m35s
```

Step 4: Cleanup

Delete the pods and namespaces to free up cluster resources.

Delete pods by namespace

```
kubectl delete pods --all -n dev
```

```
kubectl delete pods --all -n prod
```

Optional: Delete any global deployments or services

```
kubectl delete svc --all
```

```
kubectl delete deployments --all
```

```
ajith@halo:~/Devops_Lab/program9$ kubectl delete pods --all -n dev
pod "backend-pod" deleted from dev namespace
pod "frontend-pod" deleted from dev namespace

ajith@halo:~/Devops_Lab/program9$ kubectl delete pods --all -n prod
pod "backend-pod" deleted from prod namespace
pod "frontend-pod" deleted from prod namespace

ajith@halo:~/Devops_Lab/program9$ kubectl delete svc --all
service "demo-pod1" deleted from default namespace
service "hello" deleted from default namespace
service "kubernetes" deleted from default namespace
service "kubia" deleted from default namespace
service "nginx-deployment" deleted from default namespace
service "nginx-service" deleted from default namespace

ajith@halo:~/Devops_Lab/program9$ kubectl delete deployments --all
deployment.apps "demo-deploy" deleted from default namespace
deployment.apps "hello" deleted from default namespace
deployment.apps "hello2" deleted from default namespace
deployment.apps "nginx-deployment" deleted from default namespace
```